

Generation Interconnection Combined Feasibility / System Impact Study Report

For Queue Project Q16-0595

Bell Shoals 138 kV
March 27, 2018

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q16-0595. The Interconnection Customer proposes a 60.0 MW battery energy storage facility interconnecting at the Bell Shoals 138 kV substation in the Virginia area. The request was received on March 09, 2016. The study evaluated system intact and single contingency conditions using the regional planning case, including 205 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 14 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$1,995,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
GRASSMERE-RED HILL 138 KV	CLOVER-BEAR 345 KV	152.5	168	5.6
BLUFF CITY-REACH 138 KV	SCOTTSVILLE-REACH 138 KV L	130.7	229	7.5
REACH-SCOTTSVILLE 138 KV	BLUFF CITY-REACH SUBSTATION 138 KV L	118.3	188	5.5
COAL MTN-GRASSMERE SUBSTATION 138 KV Line	BEAR-COVER SUBSTATION 345 KV	111.1	257	5.6
COA MTN SUBSTATION-CENTRAL AUSTIN 138 KV	BEAR SUBSTATION-CLOVER 345 KV	110.3	266	5.6
CENTRAL AUSTIN-WAYNESBORO 138 KV	BEAR-CLOVER 345 KV L	109.4	277	5.6
BLUFF CITY-LAKE LURE 138 KV Line	FLUVANNA P.S-LOUISA GENERATION FACILITY 345 KV	107.8	480	8.4
ALDERWOOD SUBSTATION-WAYNESBORO 138 KV	BEAR-CLOVER 345 KV line	107.4	305	5.6
AIBONITO PDS-SHENANDOAH 138 KV L	BEAR-CLOVER 345 KV line	107.3	293	5.1
ROCK ISLAND DAM-SCOTTSVILLE 138 KV	BLUFF CITY-REACH 138 KV line	107.2	190	5.5
FRANKLIN-ELK GROVE-NEW LONDON 138 KV	BEAR-CLOVER SUBSTATION 345 KV L	102.5	357	14.9
MECKLENBURG POWER STATION-PERSON SUBSTATION 345 KV	BEAR-CLOVER 345 KV L	101.6	2,318	27.4

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
ALDERWOD SUBSTATION-CARSON ICE COGEN 138 KV Line	REACH-CANDELARIA 138 KV Line	101.3	233	5.0
AIBONITO PDS-S CRK 138 KV	CLOVER SUBSTATION-BEAR 345 KV Line	101.3	471	7.6

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
GRASSMERE-RED HILL 138 KV	Rebuild line with higher-capacity conductor	\$2,834,000	4% (\$119,000)
BLUFF CITY-REACH 138 KV	Rebuild line with higher-capacity conductor	\$18,436,000	2% (\$354,000)
REACH-SCOTTSVILLE 138 KV	Rebuild line with higher-capacity conductor	\$6,776,000	2% (\$152,000)
COAL MTN-GRASSMERE SUBSTATION 138 KV Line	Rebuild line with higher-capacity conductor	\$3,726,000	4% (\$156,000)
COALMTN-CENTRAL AUSTIN 138 KV	Rebuild line with higher-capacity conductor	\$3,815,000	4% (\$160,000)
CENTRAL AUSTIN-WAYNESBORO SUBSTATION 138 KV L	Reconductor line	\$947,000	4% (\$40,000)
BLUFF CITY-LAKE LURE 138 KV Line	Reconductor line	\$4,487,000	1% (\$57,000)
WAYNESBORO-ALDERWOOD 138 KV line	Reconductor line	\$1,638,000	4% (\$69,000)
AIBONITO PDS-SHENANDOAH 138 KV L	Reconductor line	\$8,299,000	4% (\$325,000)
ROCK ISLAND DAM-SCOTTVILLE 138 KV	Reconductor line	\$2,760,000	2% (\$62,000)
NEW LONDON-FRANKLIN-ELK GROVE 138 KV	Reconductor line	\$1,149,000	2% (\$29,000)
MECKLENBURG POWER STATION-PERSON 345 KV Line	Reconductor line	\$26,571,000	1% (\$351,000)
ALDERWOOD-CARSON IE COGEN 138 KV	Reconductor line	\$1,806,000	4% (\$79,000)
SOUTH CRK-AIBONITO PDS 138 KV	Reconductor line	\$2,246,000	2% (\$42,000)
	Total allocated to project		\$1,995,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.